

Gray to Binary Code converter report

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1 What is Gray to Binary and How to convert Gray to Binary ?

An Gray-to-Binary Code converter is a combinational Logic circuit that converts a Gray code into its equivalent Binary code.

1.1 For a 4bit converter:

- Input: 4-bit Binary numbers
- Output: 4-bit Gray code

2 Gray to Binary Conversion Rule

2.1 For a 4-bit Gray Code:

Gray: G3 G2 G1 G0 Binary: B3 B2 B1 B0

2.2 The Binary code is generated using XOR Operation:

1	B3 = G3 ;
2	B2 = G3 ^ G2 ;
3	B1 = G2 ^ G1 ;
4	B0 = G1 ^ G0 ;

Therefore,

1	B = G ^ (G >> 1) ;
---	--------------------

3 Implementation

3.1 Thus, the circuit requirements:

- 3x XOR gates
- 1x Direct Connection

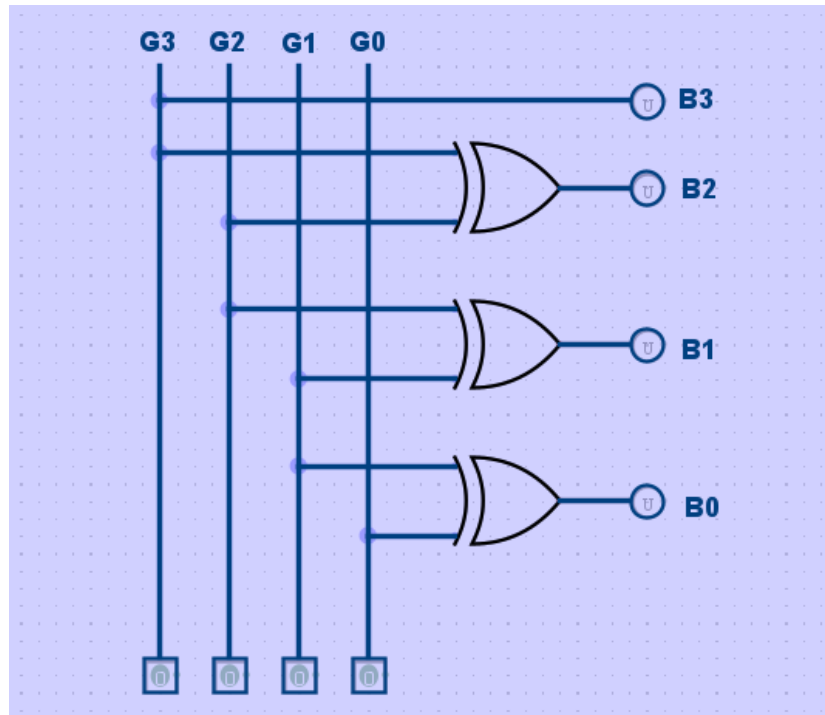


Figure 1: Gray to Binary Logic Diagram

3.2 Verilog Implementation:

```

1      module BinarytoGray(
2          input [3:0] G,
3          output [3:0] B
4      );
5
6          assign B3 = G3;
7          assign B2 = G3 ^ G2;
8          assign B1 = G2 ^ G1;
9          assign B0 = G1 ^ G0;
10
11      endmodule

```

3.3 Compact Implementation:

The same circuit can also be written using the Binary to gray equation:

```
1      content...module BinarytoGray(  
2          input wire [3:0] G,  
3          output wire [3:0] B  
4      );  
5  
6          assign B = G ^ (G >> 1);  
7  
8      endmodule
```

Both implementation produce same result.

4 Example

Gray = 0110

```
1      B3 = G3 (0) Direct Connection 0 = 0  
2      B2 = G3 ^ G2 1 ^ 0 = (1)  
3      B1 = G2 ^ G1 0 ^ 1 = (0)  
4      B0 = G1 ^ G0 1 ^ 0 = (1)
```

Therefore, Gray = 0110 Gray = 0101

5 Verification

The design can be verified using verilog testbench. The testbench applies different 4bit binary inputs and checks whether the generated gray code is correct.

5.1 Example:

6 Used Tools

- Verilog hdl
- Icarus verilog

Binary	Gray	Decimal
0000	0000	0
0001	0001	1
0010	0011	2
0011	0010	3
0100	0110	4
0101	0111	5
0110	0101	6
0111	0100	7
1000	1100	8
1001	1101	9
1010	1111	10

Table 1: The complete 4-bit truth table can be used to verified all 16 possible input combinations

- GTKWave
- VS code
- Linux/WSL

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